

XI.EXPLORE TOR NETWORK ON ANONYMOUS COMMUNICATION

Aim:

To explore and configure Tor and proxychains in order to enable anonymous communication and route traffic through the Tor network for basic penetration testing.

Procedure:

1. Install Tor: `apt-get install tor`
2. Start Tor service: `service tor start`
3. Install Proxychains: `apt install proxychains`
4. Edit config: `nano /etc/proxychains.conf`
5. Uncomment `dynamic_chain` and `proxy_dns`
6. Run application using: `proxychains firefox`
7. Verify IP and DNS changes online
8. Install and launch Tor Browser and set security to “Safe

Tasks:

Task 1: Tor

Answer the questions below

Run `apt-get install tor` to install/update your Tor packages.

No answer needed

✓ Correct Answer

Run `service tor start` to start the Tor service.

No answer needed

✓ Correct Answer

Run `service tor status` to check Tor's availability.

No answer needed

✓ Correct Answer

Run `service tor stop` to stop the Tor service.

No answer needed

✓ Correct Answer

Task 2: Proxychains

Answer the questions below

Let's start with running `apt install proxychains` to install/update proxychains tool.

No answer needed

✓ Correct Answer

Now it's time to configure proxychains to work properly.

Run `nano /etc/proxychains.conf` to edit the settings. (Note: You can use any text editing tool instead of nano.)

No answer needed

✓ Correct Answer

We can now see, that most of the methods are under comment mark. You can read their description and decide on using one of them in the future. For this lesson let's uncomment `dynamic_chain` and comment others (simply put `#` to the left). Additionally, it is useful to uncomment `proxy_dns` in order to prevent DNS leak. Scroll through the document and see whenever you want to add some additional proxies at the bottom of the page (which is not required at this point).

Apply all the settings.

No answer needed

✓ Correct Answer

Now let's check our settings.

Start the TOR service and run `proxychains firefox`. Usually, you are required to put `proxychains` command before anything in order to force it to transfer data through Tor.

No answer needed

✓ Correct Answer

After the Firefox has loaded, check if your IP address has changed with any website that provides such information. Also, try running a test on `dnsleaktest.com` and see if your DNS address changed too.

NOTE: All other web browser windows should be closed before opening Firefox through proxychains!

No answer needed

✓ Correct Answer

Task 3: Tor Browser

Answer the questions below

Finish the installation.

No answer needed

✓ Correct Answer

Launch the Tor Browser and set your privacy settings to Level 2 (Safer).

No answer needed

✓ Correct Answer

What is the search engine at the following onion address: `https://duckduckgogg42xjoc72x3sjasowoarfbgcmvfimaftt6twagswzczad.onion/`

DuckDuckGo

✓ Correct Answer

Result:

Hence Tor and proxychains were installed and configured.